[PLACEHOLDER TITLE]

Starting from the past: transient initialization and the input-history problem in soil carbon models

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Abstract

[PLACEHOLDER ABSTRACT.] One paragraph: steady-state initialization is a near-universal but rarely examined convention; an equilibrium start is already at its saturated end-state, so it cannot reproduce an accumulating sink that is only now beginning to saturate — the very dynamic observed in Finnish forest soils. We test a transient alternative across six models under one fair frame; it captures the accumulation-then-saturation as a proof of concept, but relocates the binding uncertainty onto litter inputs — especially their unknown history — pointing to historical, interdisciplinary reconstruction as the real frontier.

Contents

Part I
Opening

1 Introduction

1.1 The convention: steady state at t_0

A near-universal default in SOC modelling. State it plainly as common practice, not yet as a problem.

1.1.1 What the assumption actually claims

That the system's past equals its present equilibrium (inputs ≈ outputs at the start). Make the hidden content of the convention explicit.

1.1.1.1 Why it is attractive

Convenient, analytically clean, needs no historical forcing. The reasons it became default.

1.1.1.2 Why it is rarely examined

It is invisible precisely because it is standard. Set up the reader to recognize their own practice.

1.1.1.3 [TEMP] The convention in the Finnish inventory (sources to fold in).

- The steady-state start is the *operational* convention of the Finnish national soil-C inventory: Yasso07 is run "into a steady state", assuming "the average level of inputs and climate has remained steady over centuries" (Lehtonen et al. 2016, Geosci. Model Dev. 9:4169). $\Rightarrow$ precisely the assumption HIKET relaxes.

- It rests on the *absence of repeated stocks*: repeated soil sampling is "the most straightforward way" to see change but "too expensive" for forest soils, so dynamics were inferred indirectly (Palosuo 2008, Diss. Forestales 61).

- It was flagged early: Peltoniemi et al. (2004, Glob. Change Biol. 10:2078) modelled soil-C change over stand age from a *chronosequence* and noted soils may sit *below* equilibrium (historical slash-and-burn), and that "the long-term trend … cannot be distinguished from the measured data unless the measurements cover at least two rotations." HIKET is the successor this called for: repeated campaigns (VMI8/Biosoil/Komeetta) + a calibrated initial state. (Aleksi Lehtonen, HIKET coauthor, is acknowledged in Peltoniemi 2004 — frame the lineage collegially.)

1.1.2 The non-equilibrium reality

Real ecosystems are transient; they carry the imprint of disturbance, land-use and productivity history.

1.1.2.1 An equilibrium start is already at the end

A steady-state start sits at zero net change — by construction already at its saturated end-state. It forecloses the accumulation phase before the simulation begins. Foreshadow the capture-failure without resolving it.

1.1.3 Managed landscapes are rarely at equilibrium

Land-use and management histories keep soils in transient states; equilibrium is the exception, not the rule. Flag Finnish forest history as the concrete case that justifies the violated assumption — stated generally here, examined in detail in the Discussion.

1.2 Why this matters

1.2.1 Scientific stakes

Initialization choices propagate into every downstream projection and into model intercomparison.

1.2.2 Policy stakes: the motivating case

Finland reports its forest soil as a sink; climate policy leans on it. Introduce Finland as the instance, not the subject.

1.2.2.1 The observed dynamic

Accumulation (VMI8$\rightarrow$Biosoil), then recent hints of saturation (Komeetta, under review). A sink caught in the act of saturating. State the phenomenon the model must reproduce.

1.3 Scope and contribution (preview)

One short paragraph previewing the move: a transient initialization, tested fairly across a model ladder. Keep it a promise, not the argument.

Part II
Challenge

2 The gap and the question

2.1 The saturation an equilibrium start cannot show

The triggering insight: to reproduce a sink that is saturating, the model must first reproduce the accumulation it is the tail of — and an equilibrium start, already saturated, cannot. This is what motivated transient init.

2.1.1 Why this generalizes beyond Finland

Any non-equilibrium system started at steady state forecloses its own transient. Lift the problem off the case study.

2.1.2 Artefact or structure?

Do divergent saturation projections reflect genuine model structure, or the initialization and calibration convention? The question the fair frame will answer.

2.2 The methodological gap

No established, calibratable way to initialize a transient run when historical forcing data are absent.

2.2.1 The fixed-value trap

Treating the unknown initial state as a single fixed number (equilibrium, or a guessed value) hides a strong, untested assumption inside the result.

2.2.2 Calibrate it, do not assume it

The alternative: make the historical start a parameter with a prior and let the data and its uncertainty speak. Plant the thread; pay it off in Methods.

2.3 Objectives and hypotheses

- H1 Transient initialization lets the models reproduce the accumulation-then-saturation a steady-state start cannot. [placeholder]
- H2 Residual model differences under a fair frame are structural and modest. [placeholder]
- H3 The binding uncertainty migrates to inputs, not kinetics. [placeholder]

Part III
Action

3 Materials and methods

3.1 Study system and data

3.1.1 Plots and SOC campaigns

The calibration set is 447 permanent forest plots drawn from the Finnish National Forest Inventory (NFI) and its associated soil surveys (BioSoil/MUSTIKKA/Komeetta lineage), spanning the boreal south–north gradient. Each plot carries mineral-soil organic carbon stocks from up to three temporally separated campaigns, which serve as the calibration and validation targets: VMI8 (1985–86), BioSoil (2006), and Komeetta ($\sim$2024). The repeated stocks — two to three observations per plot separated by decades — are what make the accumulation *dynamic* observable rather than a single snapshot; this is the empirical basis for testing a transient against a steady-state start. Plots with zero modelled litter, without an organic F/H layer, or with an implied mean residence time > 100 yr are excluded. The end use is the Finnish national greenhouse-gas inventory, for which Yasso is the operational model.

3.1.2 Litter inputs

Plot-level litter is taken from the Tupek et al. dataset (Zenodo `10.5281/zenodo.19736499`), already expressed in $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ and fractionated into the AWEN quality classes required by Yasso: a non-woody component (foliage and fine roots) plus fine- and coarse-woody components (branches, coarse roots, stem bark, stumps). It is a *tree*-litter product covering the full living-tree biomass-component turnover — so belowground litter (fine *and* coarse roots) is included — but understorey/ground-vegetation litter and aboveground tree mortality (deadwood) are not represented, which becomes relevant to σ_{input} below. The population mean rises from $\sim$2.1 to a mid-2000s peak of $\sim$3.0 and then plateaus near $\sim$2.8 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$, with a strong south–north gradient across plots. This rise-then-plateau history is load-bearing twice over: it is the forcing that drives the observed accumulation-then-saturation (Results), and its level is what the input multiplier σ_{input} later interrogates (Discussion).

3.1.3 Climate forcing

Climate is gridded daily weather over 1961–2025, aggregated per plot into each model's climate modifier ξ. Because ξ is a nonlinear function of temperature (and, in Yasso15/20, precipitation), the annual modifier is evaluated year by year and then averaged — $\xi(\overline{\text{climate}}) \neq \overline{\xi_{\text{annual}}}$, a Jensen-inequality bias of 5–15% in boreal settings that we correct explicitly rather than aggregate climate first.

3.2 The model ladder

We span six models on two crossed axes — pool complexity and climate integration — so that no conclusion rests on a single structure. The roles are asymmetric by design and stated up front: **Yasso is the primary, operational model** (it is the Finnish GHG-inventory model, so it is the one that matters for the application), while **SP1/TP2/TP3 form a structural robustness**

ladder. The simpler models are not competitors for a "best" model but a sensitivity test — a check that the transient-initialization and input-uncertainty conclusions survive across model structure. This reframes the complexity axis as robustness rather than an optimal-complexity search.

3.2.1 Pool-complexity axis

Four structures of increasing pool count, all linear donor-controlled compartment systems: **SP1**, a single bulk SOC pool (one time-scale); **TP2**, an active and a humus pool in the ICBM two-pool form (fast litter turnover feeding a slow reservoir through a humification fraction); **TP3**, a three-pool active→slow→humus sequential cascade; and **Yasso07**, the five-pool AWEN+humus structure in which litter is split by chemical quality. Climbing the ladder adds internal structure while the forcing, data, and likelihood are held fixed.

3.2.2 Climate-integration axis

Three five-pool Yasso variants that share the AWEN+humus structure but differ in how climate enters the decomposition rates: **Yasso07** (temperature response of Tuomi et al. 2009), **Yasso15** (recalibrated, with an explicit precipitation term), and **Yasso20** (the Viskari et al. 2022 structure). For Yasso20 we free all twelve inter-pool transfer fractions (previously six fixed as structural zeros and three algebraically derived); the original constraints reflected a warm-site global dataset, and fixing them here would confound climate-integration differences with a calibration asymmetry.

3.3 Bayesian calibration: the enabling machinery

Each model is calibrated in a Bayesian framework (the `BayesianTools` DEzs differential-evolution sampler): five chains of 50,000 iterations each, the first 5,000 discarded as burn-in, with a fixed seed for reproducibility. The posterior combines the multiplicative-normal likelihood (below) with the homogenized priors; inference returns full joint posteriors over rates, fractions, and the two auxiliary uncertainty parameters. The role of the Bayesian frame here is enabling rather than novel: it is the machinery that turns the unknown initial state from a fixed assumption into a calibrated quantity.

3.3.1 Why a Bayesian frame is the natural fit

The initialization problem is fundamentally one of *incomplete information* about the past: we have no record of the forcing that produced the pre-observation state. A Bayesian frame is the natural fit because the prior encodes what little is known about that history and the posterior carries the residual uncertainty forward, instead of collapsing it to a single assumed value as the steady-state convention does.

3.3.2 Uncertainty that propagates, not a likelihood patch

σ_{init} is a prior on the strength of the historical (initial) carbon state, not a term added to the likelihood at the end. Its spread therefore propagates into *both* the initial condition and the forward dynamics: a wider σ_{init} legitimately widens the posterior-predictive envelope because it represents genuine ignorance about the starting point, not observation noise. This distinction — a propagated prior on the state versus a fitted error term — is what makes the transient start an inferential object rather than a numerical fix.

3.4 Transient initialization (the protagonist)

3.4.1 Pushing the equilibrium assumption back in time

Rather than assume equilibrium at the observation start t_0 (VMI8, 1985), each draw initializes the pools at their analytical steady state under a much earlier historical flux (the pre-initialization year 1917) and then runs the model *transiently* forward from 1917 to t_0. The equilibrium assumption is thus displaced ~ 70 years into the past, where it is far less consequential: by t_0 the imprint of that assumed start has largely dissipated, and the model arrives at the observation window still accumulating — the state an equilibrium start at t_0 can never occupy.

3.4.1.1 The fast/slow relaxation argument

The pre-run works because the pools relax on separated time-scales. The fast pools turn over in years and fully forget their initial value within the 68-year spin-up; only the slow humus pool, with a residence time of a century or more, still "remembers" the assumed start at t_0. Displacing the equilibrium assumption backwards therefore confines its residual influence to the one pool where it is least resolvable — and it is exactly this pool whose uncertain strength σ_{init} absorbs.

3.4.2 Calibrating the historical anchor

The pre-observation forcing is not known, so it is not fixed. The strength of the 1917 historical state enters as the calibrated parameter σ_{init} (a proportional scaling on the initial carbon state, log-transformed, prior centred at 1), which every draw infers jointly with the rest. The *shape* of the input path between 1917 and t_0 is no longer a bare linear ramp: it follows the reconstructed NFI growing-stock history, since litter scales with standing biomass — a physically grounded, near-free reconstruction of the pre-observation input trajectory that targets the coarse pre-1985 phase (see the forest-history discussion and the growing-stock figure). Full pre-run construction in Appendix C.

3.5 A fair frame for intercomparison

The design principle: any residual difference must be attributable to structure alone.

3.5.1 Shared data, likelihood, forcing, filtering

All six models see exactly the same inputs: the same 447 plots, the same litter and climate forcing, the same multiplicative-normal likelihood, the same plot-exclusion filtering, the same holdout split, and the same fixed seed. Any residual difference between models is therefore attributable to structure, not to a difference in what each model was shown.

3.5.2 Exact integration in every model

All six models integrate their linear pool system *exactly*, so that differences between them are structural rather than numerical: SP1 and TP2 use the closed-form per-step solution; TP3 uses the closed-form matrix exponential of its lower-triangular cascade generator (divided differences of the pool eigenvalues); and the Yasso variants use a Fortran matrix-exponential back-end. This matters because TP3 was previously the sole approximate integrator — an explicit forward-Euler step that rings when the active pool turns over faster than the annual step ($\alpha_A \xi > 1$), producing an interannual oscillation once mistaken for a climate response. With exact integration that numerical artefact is removed; a bounded, genuinely physical fast-pool oscillation survives (Discussion). Solvers and the discarded Euler path in Appendix E.

3.5.3 Homogenized priors: method, not number

Priors are homogenized in *construction, scale, constraint, and degrees of freedom* — not in raw value (identical numbers, tried first, broke Yasso07). A three-tier scheme applies to every model: (i) climate and size parameters take per-model centres and widths from that model's *own* published calibration, homogeneous in method and scale rather than magnitude; (ii) every inter-pool partitioning fraction — the twelve lateral transfer fractions in the Yasso family and the one or two sequential humification fractions in TP2/TP3 — shares a common weakly-informative logit prior (SD 0.4) with per-model external centres, wired by explicit per-fraction listing; and (iii) the auxiliary parameters σ_{init} and σ_{input} use an identical construction everywhere. Because the construction is common, any surviving divergence is attributable to structure and data, not to prior asymmetry. Full parameter-class × prior-criteria tables in Appendix B.

3.5.4 Degrees of freedom, set by external information

The fair frame requires not just equal degrees of freedom but equal external *information*, and the cleanest way to secure it is to parameterize every model the same way Yasso already is: *its decomposition rates are not fit to the soil-carbon data at all.* In Yasso the rate vector is held fixed at its litterbag-calibrated values (Tuomi et al. 2009/2011); what Yasso calibrates is the twelve inter-pool transfer fractions, the climate response, the woody-size modifier, and the two auxiliary uncertainty parameters. The simple models are brought onto the same footing. Their *intrinsic decomposition rates* are anchored to the Ultuna bare-fallow ICBM calibration (Andrén & Kätterer 1997; $k_1 = 0.8$, $k_2 = 0.00605$ yr^{-1}): the fast rate — litterbag-grounded and well transferable — is held fixed, while the single poorly-transferable slow rate (k_2, optimized on arable soil) carries a very-informative prior, so the data may nudge it and the posterior width reports how far a boreal forest soil departs from that anchor. Their *humification fractions*, by contrast, are the analogue of Yasso's free lateral fractions — sequential partitioning of outflow into the next, slower pool — and are therefore left *free* under the same common logit prior, re-centred on the ICBM humification value. Fixing them too would leave the simple models with less partitioning freedom than Yasso, not more homogeneity. How completely a structure's rates can be anchored is set by whether its time-scale structure matches an existing calibration: TP2 (two time-scales) maps onto ICBM one-to-one; SP1 (one time-scale) inherits the ICBM bulk residence time; and TP3 (three time-scales) places both of its slow pools at the ICBM "old" subsystem scale, its fast pool at the young scale. The result is uniform across all six models: *no model fits its kinetics to the SOC data*, and every model spends its calibrated freedom on the same four things — the partitioning fractions, the climate response, the litter input, and the initial state. Climbing the pool-complexity ladder therefore adds only free partitioning degrees of freedom (none in SP1, one in TP2, two in TP3, twelve in the Yasso family); identifiability then tracks information, by design rather than by accident. [strand] This convention is also what makes the stratified extension Yasso-only: with rates anchored, Yasso's stratifiable knobs are inputs + the humus-formation fraction — see the optional strand in the Outlook.

3.6 Calibration and inference

3.6.1 Error model

The likelihood is multiplicative-normal (proportional error): the observation standard deviation scales with the predicted stock rather than being constant. This is the appropriate form for SOC stocks, which span an order of magnitude across plots, and it operates on the log scale where residuals are near-Gaussian. One consequence is an asymmetric penalisation of under- versus over-prediction, which leaves a small systematic offset in the fitted bias; we flag it here and return to it in the Discussion, where the $\sim 11\%$ over-prediction is shown to be a property of this error model rather than of the biogeochemistry.

3.6.2 Physically-bounded input priors

Left as a scale-free multiplier centred at one, σ_{input} can drag the effective litter flux to physically impossible levels (the unbounded prior let two models reach 13–20× boreal net primary production). We therefore bound the *effective flux* — $\sigma_{\text{input}} \times J$, and the coupled 1917 pre-run flux — to a boreal NPP envelope ($\sim$ 0.5–8.7 $\text{tC ha}^{-1}\,\text{yr}^{-1}$), homogeneous across models, via a bounded re-parameterisation of the $\sigma_{\text{init}}/\sigma_{\text{input}}$ pair (a scaled-logit "flux-pair" transform) rather than a hard truncation, so the sampler still mixes across the boundary. In addition, because the litter product, though it includes belowground (root) litter, structurally omits understorey/ground-vegetation and mycorrhizal inputs (and aboveground tree mortality), the σ_{input} prior is centred on the physically expected correction (> 1) rather than on one, so its posterior is read against expected total litter rather than against the tree-only baseline. Rationale, envelope and mechanism in Appendix A.

3.6.3 Sampler and diagnostics

Sampling uses the DEzs differential-evolution sampler (five chains, 50,000 iterations, 5,000 burn-in). Mixing is checked by the Gelman–Rubin $\hat{R}$ (target < 1.05) and effective sample size, and the proposal infinity-rate is monitored to catch prior-driven forward blow-ups. We distinguish mixing from *identifiability*: with healthy mixing the sampler traverses the non-identified ridges cleanly, so a fraction can show good $\hat{R}$ yet remain data-uninformed. The genuine non-identifiability signature is therefore read elsewhere — prior-pinning, near-perfect same-pool anti-correlation ridges, and low KL divergence of posterior from prior (< 1 nat) — not from poor $\hat{R}$.

3.6.4 Holdout split and validation

A fixed subset of plots is held out of the likelihood entirely and used only for out-of-sample evaluation. Predictive skill (R^2, RMSE, bias) is reported separately for the calibration and holdout sets, so that in-sample fit and genuine predictive performance are never conflated — the distinction that later separates in-sample skill from the forecast divergence.

3.6.5 Software, compute and reproducibility

The pipeline is implemented in R, with the Yasso decomposition solved by a compiled Fortran matrix-exponential back-end and the simple models solved in closed form. All stochastic steps use a fixed seed (`set.seed(2025)`), and the production calibration runs on an HPC cluster (one job per model). Code is version-controlled and the final outputs are archived openly; per-model posterior and predictive bundles are keyed by run identifier so that each predictive run is traceable to its calibration.

3.7 Predictive and residual analysis

For each model we draw posterior-predictive SOC trajectories (an ensemble of parameter draws pushed through the forward model, initial state included), from which the campaign-year stocks, skill metrics, and forecast envelopes are taken. Plot-level residuals are then regressed against candidate site covariates (soil chemistry, texture, climate summaries, stand basal area, litter quality) by random forest, to locate where recoverable signal remains after calibration. Across models the leading, shared driver is stand basal area — a productivity and input proxy, not a kinetic variable — pointing the refinement toward better inputs rather than more pool structure.

4 Results

4.1 Capturing accumulation-then-saturation

The headline: transiently-initialized models reproduce the rise and its incipient bend-over; the equilibrium-started flat/declining shape was the start, not the structure.

4.1.1 Trajectory shape vs. level

Shape change, not just offset. [placeholder figure]

4.1.2 Match to campaign means

2006 and 2024 means tracked. [placeholder]

4.2 Structural differences under the fair frame

Real but modest skill spread across the six. [placeholder table]

4.2.1 Where simple beats complex

TP2/TP3 track shape best; stiffer Yasso family worst. Brief.

4.3 What the data can and cannot constrain

Set up the Discussion: kinetics weakly constrained; inputs do the work.

4.3.1 Convergence vs. identifiability

Good R-hat coexisting with non-identified fractions. State the apparent paradox, resolve in Discussion.

4.3.2 Input multiplier behaviour

σ_{input} does different work in different models. The hinge result.

4.4 Residual structure

RF importance; stand basal area as the leading, diffuse, shared driver. [placeholder figure]

Part IV
Resolution

5 Discussion

5.1 Capturing saturation requires remembering the accumulation

Resolve the Challenge: the difficulty was the equilibrium start, not the soil; a transient start lets every model represent the observed dynamic. The central payoff.

5.1.1 Proof of concept, not a verdict on Finland

What the result does and does not license. Calibrate the claim.

5.1.2 Generalization to the model class

Wherever steady-state init meets a non-equilibrium system, it forecloses the transient the data demand. Lift off Finland.

5.2 The binding uncertainty is the inputs (the σ_{input} reveal)

The data cannot resolve kinetics; the input multiplier absorbs what structure cannot. So the frontier is inputs, not soil chemistry.

5.2.1 The multiplier crosses a physical line (quantitative payoff)

The physical bound turns this into a clean counterfactual, and that counterfactual is the cleanest proof that inputs — not kinetics — were the lever. Rescored against the earlier *unbounded* calibration, the ceiling costs predictive skill for *only* the two runaway models (TP2, TP3), the ones that had reached 34–50 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ of manufactured litter; SP1 and the three Yassos, already inside the envelope, are unchanged. Had the good fit of TP2/TP3 rested on structure, bounding a *shared* input nuisance could not have singled those two models out. The *selectivity* of the skill drop is therefore evidence that the surplus fit was bought with unphysical carbon. We frame this as *consistent with* input/forcing error, not proof: the global design cannot exclude local kinetic variation (see the excursus and the optional stratified strand). Before/after diagnostic: Appendix A, Fig. 1.

5.2.2 Historical inputs: the unrecoverable past

No forcing data pre-observation; σ_{init} stands in for an entire history, held in the slow pool that remembers it. The deepest uncertainty.

5.2.3 Belowground inputs and other ramifications

Root/rhizodeposition inputs poorly known; a further axis of the same problem.

5.2.3.1 [TEMP] Missing understorey litter (why σ_{input} ¿ 1 is expected).

- The litter product used here is *tree litter* — it covers the full living-tree biomass-component turnover, so *roots (fine and coarse) are included* (confirmed by B. Tupek and the dataset size-class table: `nwl`=foliage+fine roots, `fwl`=branches+coarse roots+stem bark, `cwl`=stumps).

What is *not* represented, in order of magnitude: understorey/ground-vegetation (incl. moss) litter (dominant); mycorrhizal mycelial turnover/exudates (large, uncertain); and aboveground tree mortality (deadwood), a genuine omission but small in managed forest. So a multiplier above one is a physical expectation, not manufactured carbon — hence the prior is re-centred at ∼1.3 (understorey-dominated) rather than 1.

- Grounding: Lehtonen et al. (2016) found understorey litter *underestimated* in Yasso07, especially in the north (bryophytes 20–60% of NDVI at high latitude); Finnish litterfall studies put understorey at ∼15% (south) to ∼33% (north) of aboveground litter. Peltoniemi et al. (2004) give component turnover rates for dwarf shrubs/bryophytes/herbs/lichen (S. Finland).
- Caveat: the fraction rises strongly S→N, so a single national multiplier cannot capture it — an argument for stratifying inputs by site class (future work).

5.3 A short excursus: why this class of models is non-identifiable

5.3.1 The structural argument

First-order kinetics (and richer topologies) let parameters interact so the data constrain net fluxes out of a pool, not the individual splits.

5.3.1.1 Sketch

For a linear pool system the observable stocks depend on the parameters only through lumped combinations; distinct parameter sets give identical predictions (parameter-space ridges). Two equations of placeholder math. Full argument in Appendix D. [placeholder]

5.3.1.2 Signature in practice

Prior-pinning, near-perfect anti-correlation ridges between same-pool fractions, low information gain (KL of posterior vs prior). The Bayesian frame is what lets us see this rather than hide it in a fitted point. Tie to Results.

5.3.1.3 Why it is not a defect to be "fixed"

It is intrinsic to the model class and the data content (few observations per plot); it bounds what calibration can ever claim.

5.3.1.4 Managing it by convention, transparently

We do not claim to resolve the rate↔fraction equifinality — we MANAGE it: where authoritative rate values exist (Yasso) we pin the chemical axis and calibrate the partitioning; where they do not (simple models) we calibrate and let the equifinality show. An explicit convention, not an identification claim — and the candour is what keeps this section credible rather than defensive. Ties back to "degrees of freedom set by external information" in Methods.

5.4 Structure is a weak lever in-sample, but it dominates the forecast

Within the calibrated window the six structures are near-interchangeable: complexity buys no skill (Results), and the excursus just showed why — the surplus degrees of freedom are non-identified, with the inputs absorbing the misfit. Extrapolated, the picture inverts. Run forward to 2084 the models agree where data constrain them and then fan out sharply — the single-pool SP1 climbs without bound ($\sim$133 $\mathrm{tC\,ha^{-1}}$), while the stiffer Yasso structures plateau ($\sim$108). The very structural choices the data could not pin down now govern the projection. In-sample skill is blind to this: it is scored only on the current stocks, which every model reproduces about equally well.

5.4.1 A cautious read, not a ranking

Process plausibility — not fit — gives a soft steer. Soils saturate, so SP1's unbounded single-pool rise is the least physically credible trajectory, and the saturating structures sit better with what we expect of a maturing sink. But the steer is soft and we keep its counter-caveat explicit: under the fair frame Yasso carries the most free degrees of freedom, so its plateau could be input-bound humus-parking rather than a better-resolved dynamic. Present data cannot adjudicate between these readings.

5.4.2 The honest output is the spread

Because the divergence is real, decision-relevant, and unfalsifiable from present data, the defensible product is not a chosen model but the *ensemble spread* — reported, not hidden. This is the deepest reason the national maps carry a between-model uncertainty panel beside the mean: the spatial SD is the forecast disagreement made geographic. And the projection a greenhouse-gas inventory most needs — multi-decade SOC change — is precisely the quantity in-sample skill cannot validate, which is why the honest hand-off to policy is the spread, not a single number. Narrowing that spread will not come from more SOC campaigns — two or three stocks per plot cannot arbitrate multi-decade dynamics — but from evidence that constrains the *trajectory*: long-term chronosequences and re-measured permanent plots, soil-warming and litter-manipulation experiments, and process-based limits on how much carbon a maturing boreal soil can hold. Until such constraints exist, the divergence is a genuine, reportable uncertainty, not a modelling failure.

5.5 The over-prediction is a property of the likelihood

Slight, systematic over-prediction follows from the multiplicative-normal error model, not from biogeochemistry. Only the bias is likelihood-sensitive; rankings are not.

5.6 Limitations

5.6.1 The coarse pre-observation phase

Linear litter interpolation under constant climate; high, deliberate uncertainty.

5.6.2 The flat-vs-rising litter tension

Plot-level inputs vs. national reconstructions; present data cannot settle it. State openly.

5.6.3 [TEMP] Between-campaign stock consistency (VMI8 vs Biosoil)

- The apparent VMI8→Biosoil sink ($\sim$184 $\mathrm{gC\,m^{-2}\,yr^{-1}}$) is implausibly large; the paired 1985–2006 plots show the *organic* layer stable but *mineral* soil C nearly doubling (incl. deep 20–40 cm) $\Rightarrow$ a between-campaign mineral-soil measurement inconsistency, not accumulation.
- Independent support: Peltoniemi et al. (2004) found *mineral soil C does not change with stand age* (only the organic layer does), and a careful total of 6.8 $\mathrm{kgC\,m^{-2}}$ to 1 m ($\approx$ our VMI8 $\sim$63 $\mathrm{tC\,ha^{-1}}$; makes Biosoil/Komeetta $\sim$100+ look high). Cite for this caveat.

- Handling (Methods): the 1985 stocks are *down-weighted* via inflated observation variance ($\sim 2\times$), not dropped; authoritative QC is ongoing (LUKE). Consequence: the "+26 $\mathrm{tC\,ha^{-1}}$ 1985 over-prediction" is likely a low VMI8 measurement, not a model bias.

5.7 A managed landscape out of equilibrium: Finnish forest history

The concrete history behind the violated equilibrium assumption: why Finnish forest soils were accumulating, not at steady state, across the observation period. Pays off the Intro's general statement.

5.7.1 From exploitation to managed growth

A sketch of the management eras — heavy historical use, then regeneration and intensified management, rising growing stock through the 20th century. [placeholder — sources to add]

5.7.2 The imprint on litter inputs and stocks

Rising biomass and shifting management make the input history non-stationary — precisely the signal an equilibrium start erases. Connect to the σ_{input} reveal and the flat-vs-rising litter tension.

5.7.3 Why this is the rule, not a Finnish peculiarity

Most managed and recovering landscapes carry comparable histories; the lesson generalizes. Tie back to the Intro's general statement.

5.8 Outlook: toward a historical perspective on inputs

The forward call. The real next step is reconstructing historical (and belowground) input regimes.

5.8.1 From diagnosis to method (now folded into M&M)

The bound itself has moved to Methods; the remaining frontier is the input *history* (temporal) and belowground inputs. See Appendix A.

5.8.2 An interdisciplinary, source-diverse effort

Reconstruction needs evidence beyond the usual datasets — grey literature and non-traditional historical sources. Frame as a research program.

5.8.3 Initialization as a calibratable, evidence-based step

Replace the equilibrium default with priors grounded in reconstructed history, calibrated under uncertainty. The thread closes here: better history → sharper prior → a start that is assumed less and inferred more. The generalizable recommendation.

5.8.3.1 Concrete near-term step (to explore next session).

The transient-init pre-run currently ramps litter *linearly* from the 1917 anchor to 1985. A near-term improvement is to use the *shape* of the NFI growing-stock history (§[History]) as that interpolation function instead: since litter scales with standing biomass, the growing-stock recovery curve is a physically grounded, essentially cost-free reconstruction of the pre-observation input path, and it targets the coarse pre-1985 phase directly.

5.9 [Optional strand] Locally stratified calibration (Yasso only)

A droppable extension — include only if data + time allow; the storyline stands without it. It is the SPATIAL counterpart of the historical-reconstruction prong: both add structure to the *forcing*, never to the kinetics.

5.9.1 Scope — the operational model only

Stratify Yasso (the operational model); SP1/TP2/TP3 stay global robustness anchors. Consistent with the rate convention: Yasso already fixes its published rates, so its free knobs (inputs + fractions) are the ones available to stratify. Triggered only "if complexity emerges as needed", not by default.

5.9.2 What to stratify — inputs and the humus-formation fraction

The physically-bounded input σ and the humus-formation fraction (the stabilisation control) — NOT the rates (fixed) nor the inter-pool splits (non-identified). Both carry external, class-varying referents; humus formation is a near-net stabilisation flux the total stock genuinely constrains, so of the fractions it is the one worth stratifying.

5.9.3 RF-guided selection, holdout-evaluated

Use the residual random forest to CHOOSE the stratifying classes (its OOB R^2 is the ceiling of covariate-recoverable local structure); then measure the stratified model's gain on the HELD-OUT plots — non-negotiable, or selecting and scoring on the same data makes the improvement circular. Report "RF says up to $X\%$ is recoverable; the stratified mechanistic model recovers $Y\%$ on holdout."

5.9.4 Hierarchical, predictive-robustness framing

Partial pooling: data-poor strata shrink to the global value (no minimum-N problem); the between-stratum variance is itself a result. Headline = improved, more ROBUST posterior predictions even where stratum parameters stay weakly identified — identifiability and predictive skill are distinct, and robustness is what the GHG-inventory use needs.

5.9.5 Equifinality as a measured quantity

Bounded inputs vs. free humus formation compete for the local variance; the physical bound PARTIALLY breaks their equifinality (one competitor has walls, the other does not), so the learned variances softly attribute local variance to input vs. stabilisation origin — cross-checked by the RF driver identity (productivity proxies $\rightarrow$ input; soil-chemistry proxies $\rightarrow$ stabilisation). Frame as "consistent with", not proof. Expect modest holdout gains against a low predictability ceiling — modest-but-robust-and-honest beats an overclaimed jump.

6 Conclusions

6.1 The experiment succeeded

Transient, calibrated initialization let a model ladder reproduce the observed accumulation and its incipient saturation — the dynamic an equilibrium start forecloses. State the proof of concept plainly.

6.2 The Bayesian approach made it possible

Treating the historical start as a calibrated parameter, with uncertainty propagated into state and dynamics, is what turned the idea into a working implementation. Credit the machinery without overclaiming novelty.

6.3 Refinement needs historical data

The approach can go much further once the pre-observation history is reconstructed from data rather than left to calibration — the open frontier, and the natural successor study. Close here.

Part V
Appendices

A Physically-bounded litter-input priors

A.0.0.1 Problem.

The input multiplier σ_{input} scales the litter forcing every model integrates. Left as a weak, scale-free prior it lets the likelihood drag two models (TP2, TP3) to effective litter fluxes of 34–50 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ — several times the net primary production of any boreal forest. The multiplier's distance from unity is therefore a *structural-adequacy gauge*, interpretable only once the prior encodes a physical ceiling.

A.0.0.2 Physical envelope.

The effective flux is $\sigma_{\mathrm{input}} \times J$, with J the shared tree-litter object (median ≈ 2.8 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$). We bound it to the boreal total-NPP envelope of Gower et al. (2001), $[0.05, 8.7]$ $\mathrm{tC\,ha^{-1}\,yr^{-1}}$, homogeneous across all six models and applied to *both* ends of the pre-run: the contemporary flux $F_{\mathrm{now}} = \sigma_{\mathrm{input}} J$ and the 1917 historical flux $F_{1917} = \sigma_{\mathrm{init}}\sigma_{\mathrm{input}} J$. The ceiling (8.7, the Gower maximum) is the binding limit; the floor is relaxed below the Gower minimum to a small, strictly-positive, non-binding guardrail — kept > 0 so that zero flux (hence zero predicted SOC and a $-\infty$ likelihood) cannot occur, and set low to pre-position for a future stratified σ. Because σ_{init} has no independent physical referent — it is a *ratio* of two fluxes — it is not bounded on its own; instead F_{1917} is bounded to the same envelope and $\sigma_{\mathrm{init}} = F_{1917}/F_{\mathrm{now}}$ is recovered. The multipliers are scalar (network-average), so local disturbance (clear-cuts) lives in the per-plot J, not in the bound.

A.0.0.3 Mechanism: bounded re-parameterisation.

A hard truncation would mix poorly precisely where it bites, since the two binding models sit against the ceiling. Instead we sample an unconstrained coordinate z and map it through a scaled logistic onto the window $[a, b]$,

$$p = g(z) = a + (b - a)\,\sigma(z), \qquad \sigma(z) = \tfrac{1}{1+e^{-z}}, \qquad z = g^{-1}(p) = \log \tfrac{p-a}{b-p},$$

so every proposal is feasible by construction and the boundary is approached asymptotically, never struck. The change of variables contributes a Jacobian term to the log-prior,

$$\log\bigl|g'(z)\bigr| = \log(p - a) + \log(b - p) - \log(b - a),$$

which keeps the physical-space prior the one intended. The two fluxes are two such coordinates; the multipliers are recovered as $\sigma_{\text{input}} = F_{\text{now}}/\bar{J}$ and $\sigma_{\text{init}} = F_{1917}/F_{\text{now}}$, with $\bar{J}$ a fixed mean-litter units bridge. Implemented as one new transform type in the calibration engine, homogeneous across the six prior specifications.

A.0.0.4 Consequence.

The escape hatch closes: misfit is forced onto the (otherwise prior-pinned) kinetics or into a visible R^2 hit, exposing structural inadequacy instead of hiding it in manufactured carbon.

A.0.0.5 Before/after diagnostic.

The main text reports only the final, flux-bounded calibration. For completeness, Figure 1 contrasts it against the earlier *unbounded* homogenized run (the calibration that motivated the bound). Without the physical ceiling, TP2 and TP3 reach median effective fluxes of 34 and 50 $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ — 13–20× boreal NPP, physically impossible — while SP1, Yasso07/15/20 already sit inside the envelope. Under the bound all six collapse into [0.9, 6.3], and, tellingly, *only* the two runaway models lose predictive skill: confirmation that they had been buying fit by manufacturing carbon, not by resolving kinetics. The unbounded run is retained only as this diagnostic (git-tagged, not a maintained parallel pipeline); it is not part of the reported results.

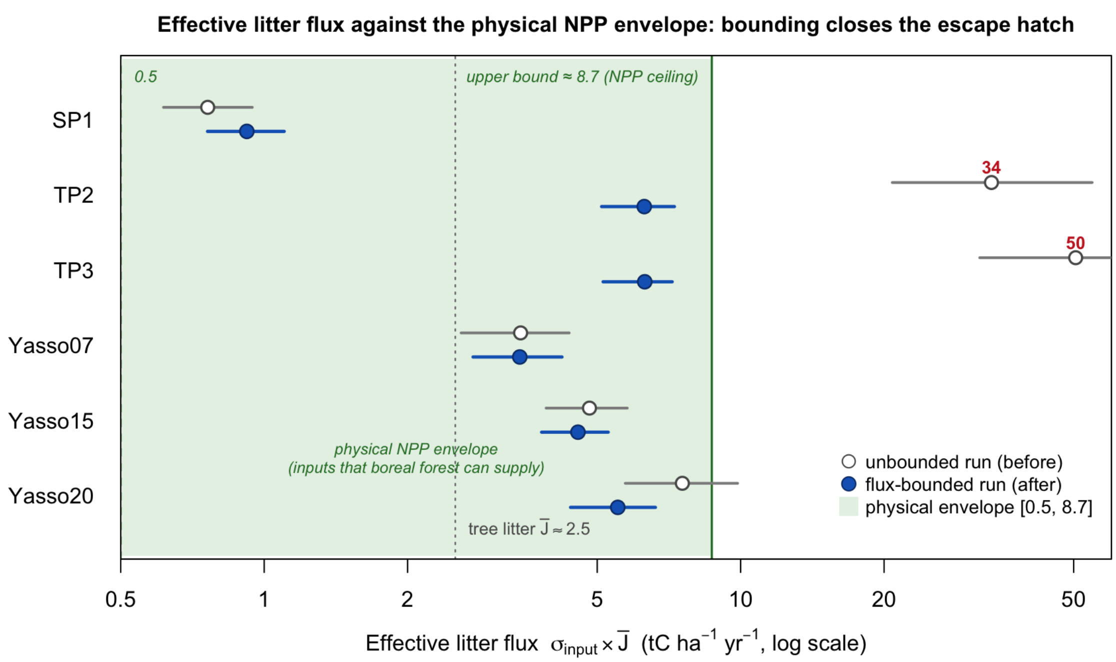

Figure 1: Effective litter flux $\sigma_{\text{input}} \times \bar{J}$ per model, on a log scale, against the physical NPP envelope [0.5, 8.7] $\mathrm{tC\,ha^{-1}\,yr^{-1}}$ (green). Grey open points = the unbounded homogenized run (before); blue filled = the flux-bounded run (after); points are posterior medians, whiskers the 95% interval. TP2/TP3 leave the envelope entirely when unbounded (34/50) and return inside it under the bound.

Full derivation, per-model diagnostics, literature envelope and references: standalone note `sigma_input_physical_bounds_note`.

B Prior homogenization: the tier scheme

[PLACEHOLDER.] The parameter-class × prior-criteria table (centre source, width source, transform/constraint, cross-model consistency) for the three tiers: climate & size, transfer fractions, auxiliary uncertainty parameters. Homogeneous in construction, scale and degrees of freedom — not in raw numbers.

B.1 Tier 1 — climate and size parameters

[placeholder table]

B.2 Tier 2 — transfer fractions

[placeholder table]

B.3 Tier 3 — auxiliary uncertainty parameters

[placeholder; σ_{init}, σ_{input} — see Appendix A for the physical bound.]

C Transient initialization: the pre-run

[PLACEHOLDER.] The 1917→1985 pre-run: analytical steady state in the past, run transiently forward to t_0. Litter interpolation, constant-ξ climate during the pre-run, the fast/slow relaxation argument, and the σ_{init} anchor.

C.1 Endpoint construction and interpolation

[placeholder equations]

C.2 Fast/slow relaxation and residual memory

[placeholder]

D Non-identifiability of the linear pool class

[PLACEHOLDER.] Why the data constrain net fluxes out of a pool, not the individual splits: lumped combinations of parameters, parameter-space ridges, and the practical signature (prior-pinning, same-pool anti-correlation, low KL of posterior vs prior).

D.1 The structural argument

[placeholder equations]

D.2 Diagnostic signature

[placeholder]

E Exact integration of the pool systems

[PLACEHOLDER.] All six models integrate their linear pool system exactly: closed-form per-step (SP1, TP2), matrix exponential of the cascade generator (TP3), Fortran matrix exponential (Yasso07/15/20). Contrast with the discarded explicit-Euler path and why numerical ringing was an artefact, not structure.

E.1 Closed-form and matrix-exponential solvers

[placeholder equations]

F Reconciling the SOC-change diagnostics

Two diagnostics of the same six calibrated trajectories can appear to tell different stories, and we reconcile them here explicitly. An earlier version of the increment figure plotted the *cumulative-average* accumulation rate,

$$\bar{r}(t) \;=\; \frac{\mathrm{SOC}(t) - \mathrm{SOC}(t_0)}{t - t_0}, \qquad t_0 = 1985,$$

whose curves *decline* over time for every model. Read too quickly, a falling curve suggests SOC is decreasing — which contradicts the absolute-stock figures (the mean-SOC trajectory and the initialization figure), where every model's mean SOC *rises* toward saturation. There is no contradiction: the two are the same data under different transforms.

Figure 2 shows the identical six trajectories through three lenses. Panel (a) is the absolute stock; it rises and bends toward saturation. Panel (b) is $\bar{r}(t)$ — the earlier figure's quantity; it declines because the *time-averaged* rate of a decelerating-but-still-rising stock necessarily falls as the saturating tail is averaged in, not because any carbon is lost. Panel (c) is the instantaneous increment $\mathrm{d\,SOC}/\mathrm{d}t$; it too declines, dipping slightly negative for the multi-pool models near saturation — the derivative view of the same bend. All three panels are mutually consistent: (b)'s decline is (a)'s curvature and (c)'s deceleration.

Panel (b) also carries a substantive reading that the absolute-stock view makes less obvious: every model's cumulative-average rate settles *below* the observed reference (dashed). Because the models over-predict the 1985 state by $\sim 26\,\mathrm{tC\,ha^{-1}}$ (the main-text initialization figure), their accumulation *since* 1985 is smaller than the observed $63 \to 105$ climb, so their time-averaged rate cannot reach the observed value. The gap between the model curves and the dashed line in panel (b) is thus a second view of the same initialization over-prediction, not an independent discrepancy.

Notes for collapsing (remove before submission)

- **Parts** = OCAR acts; will dissolve into standard Intro/Methods/Results/Discussion at cleanup.
- Many heading levels are intentional scaffolding; merge aggressively once the story is fixed.
- Keep the protagonist (initialization problem) constant; do not let σ_{input} or non-identifiability take over the spine.

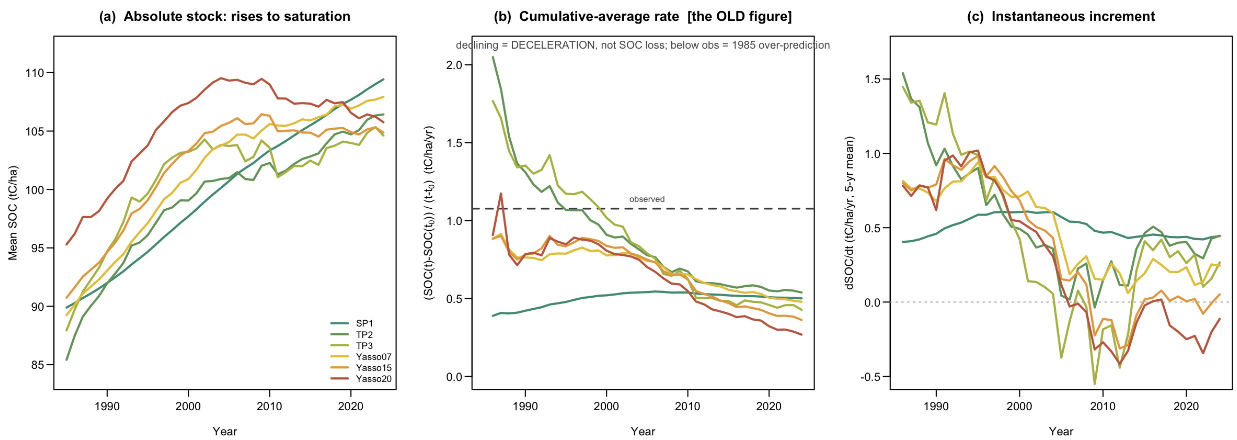

Figure 2: The same six calibrated trajectories under three transforms. (a) absolute mean SOC; (b) the cumulative-average rate $(\mathrm{SOC}(t) - \mathrm{SOC}(t_0))/(t - t_0)$ used in the earlier increment figure; (c) the instantaneous annual increment (5-year mean). The decline in (b) is the deceleration of a still-rising stock, not a loss of carbon.